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COMPSCI 160  
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#### Project Milestone 4: Implementation

**We have already submitted the Google Form to report AI coding tool usage.**

**Project URL:** <https://sane24.github.io/Anchor-App>

**GitHub Repository URL:** <https://github.com/Sane24/Anchor-App>

#### Feature Testing:

| Feature tested                                                        | Expected result                                                                                                                                                                                                           | Actual result                                                                                                                                                                       | Fix / iteration                                                                                                                                                                             |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sign up, log in, and log out                                          | A new user can create an account with name, email, and password, gets signed in automatically, sees their identity reflected in Settings (Account section), can sign out, and can sign back in with the same credentials. | User can create account and sign in to existing account                                                                                                                             | <b>Iteration:</b> added password requirements for signing in (password must be at least 8 characters and include an uppercase letter, a number, and a symbol) and forgot password retrieval |
| Navigate between Today, Garden, This Week, Reflection, Settings pages | Can tap each tab to load the screen without errors. Navigating back doesn't lose state unexpectedly                                                                                                                       | Successful navigation between tabs through the home navigation bar. Opening Briefing (from Today) and Timer (from an anchor) work correctly even though they're not tabs themselves | <b>Iteration:</b> made the Anchor logo clickable                                                                                                                                            |
| Resume progress after refreshing or                                   | Change the input of the first step on Today's screen, and can still see the changed results after refreshing the page                                                                                                     | The change is not saved after refreshing, would revert                                                                                                                              | <b>Fixed:</b> wired the input's onChange to save directly to the shared plan store on every edit, so the value is already                                                                   |

|                                                                                     |                                                                                                            |                                                                                                    |                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| logging back in                                                                     |                                                                                                            |                                                                                                    | persisted by the time a refresh happens                                                                                                                                                                                                                                  |
| Confirm that one user cannot access another user's progress                         | Each user's tasks and progress are private and not visible to other accounts                               | Successful: Verified on a separate browser/account, so no tasks from the original account appeared | No need to fix: Supabase's row-level security policies already scope all reads/writes to auth.uid()                                                                                                                                                                      |
| Morning briefing: review items and pick top 3 tasks from calendar to Today's Anchor | Can successfully choose the preferred 3 tasks from the calendar to Today's Anchor screen                   | Selected tasks did not appear on Today                                                             | Briefing's save action wrote to localStorage under a key nothing else was reading from.<br><b>Iteration:</b> connected Briefing's save to the shared store.js plan functions so Today reads the same saved anchors.                                                      |
| Focus timer with popup settings                                                     | Can turn off the music without going back to the focus timer screen                                        | Music kept playing even while the timer was paused                                                 | <b>Fixed:</b> the pause button only stopped the countdown, not the separate audio element.<br>Iteration: added an explicit audio .pause() call inside the same pause handler that stops the timer.                                                                       |
| Tomorrow: Postpone the task                                                         | Can see the task set to do tomorrow popping on tomorrow's calendar                                         | No change appeared. The task did not carry over                                                    | Root cause: the postpone action updated the task's date locally but never re-saved it to the store, so the change was lost.<br>Iteration: postpone now updates the task's date field in the store directly, so it appears under the new day everywhere the data is read. |
| Change timer duration                                                               | Predetermined times (5, 15, 25 minutes) can be set by the user and automatically count down when the start | Successful, all buttons are functional and can edit the timer duration.                            | <b>Iteration:</b> sliding the timer circle adjusts the countdown length, allowing for manual input by the user.                                                                                                                                                          |

|                                                               |                                                                                                                                          |                                                                                   |                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                               | button is clicked. +5 or -5 buttons can also be used to edit the timer duration.                                                         |                                                                                   |                                                                                                                                                                                                                                                                                                                                                     |
| Connect Google account                                        | Clicking “Connect” opens a Google will open a sign-in popup; briefings fetch real data                                                   | Access blocked for emails that were not added to the Test Users in Google console | OAuth app is in Testing publishing status, which restricts sign-in to an explicit allowlist.<br>Iteration: added required teammate/tester emails to the Test Users list in Google Cloud Console.<br>Known limitation: this is not a permanent fix, any new user still needs to be manually added unless the app is moved to Production or verified. |
| Manually add tasks                                            | Can add tasks to Today’s Anchor by typing. Manually added task selected as an Anchor                                                     | Input field was not fully implemented                                             | Root cause: no input field existed yet for manual task entry.<br>Iteration: added a text input and “Add” button that appends a new task to the list on submit.                                                                                                                                                                                      |
| Enable notification reminders and receive one at the set time | Set morning reminder 2 min ahead, switch to another tab, notification fires at the set time even when not on Settings                    | Successful, worked as expected                                                    | No fix needed.                                                                                                                                                                                                                                                                                                                                      |
| This Week day filtering                                       | Tapping a day chip shows only that day’s tasks and events, with correct due times and source tags (Gmail/Calendar/manually added by you) | Successful, worked as expected                                                    | No fix needed.                                                                                                                                                                                                                                                                                                                                      |

### Stress testing:

| Test | What happened | What changed afterward |
|------|---------------|------------------------|
|------|---------------|------------------------|

|                                                                                                                                    |                                                                                                                         |                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Submitted the task input with no text entered                                                                                      | The Add button correctly stayed disabled/inactive, preventing an empty task from being created                          | No change needed — existing validation already blocks this case                                                                                                     |
| Entered a very long string (100+ characters) into the task input field                                                             | Text overflowed its container and spilled off the visible screen edge                                                   | Added min-width: 0 + overflow-wrap: anywhere; retested 8 screens, 0 overflow                                                                                        |
| Refreshed the browser while the focus timer was actively running                                                                   | The timer reset to its starting duration, losing the in-progress countdown                                              | Added logic to persist the timer's remaining time and running state to storage on each tick, so a refresh restores the timer where it left off instead of resetting |
| Rapidly clicked the Start button on the timer multiple times in succession                                                         | The timer started normally once; repeated clicks had no unintended effect (no duplicate timers, no double-counted time) | No changes needed, the start handler is already idempotent while a session is active                                                                                |
| Loaded the app on a very small screen width (below 360px)                                                                          | The layout remained functional and legible, with no broken or clipped elements.                                         | No changes needed. Responsive styles already handle this width                                                                                                      |
| Manually edited saved notification settings in localStorage into invalid JSON, then reloaded the app                               | The app detected the corrupted data and fell back to default notification settings instead of crashing                  | No changes needed, error handling covers malformed stored data                                                                                                      |
| Forced an invalid reminder time value ("99:99") into settings and triggered the scheduler                                          | The scheduler did not throw an error or freeze the app; it silently ignored the invalid time                            | No change needed. Input was cleaned                                                                                                                                 |
| Added a large data volume by injecting 500 tasks through the data layer, today and this week should still load and stay responsive | Today and This Week screens still loaded correctly and remained responsive, with no noticeable lag                      | No change needed. Current rendering approach scales to this volume without optimization                                                                             |
| Created a task with a missing due time field                                                                                       | The task rendered correctly in the day view without crashing the app or the storage layer                               | No change needed. Missing fields are already handled with safe backup plan                                                                                          |

Checked whether tasks land on the correct calendar day across different user timezones

Bug found: sample task dates are generated in UTC, while This Week keys days using local time. For users in UTC+ timezones (e.g., Europe, Asia), tasks appeared on the wrong calendar day. The bug was not caught earlier because all testing had been done in US timezones, where the UTC/local mismatch doesn't cross a day boundary.

Identified as an open bug, not yet fixed. Planned fix for next iteration: generate day-offset dates using local date formatting (matching the same toKey function This Week already uses) instead of UTC, so date generation and date lookup use a consistent timezone basis.

## Updated Development Plan

Sheets:  Anchor Implementation Plan

| Anchor Development Plan, Sun 7/26 to Fri 7/31                                                                                                                                                                                                                               |                                                                                                                                        | Who                                                                                                                                                                                                                                                                                                                                                                                   | Hours | Dependencies                                                                                                                                                                            | It's done when                                                                       | Status      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------|
| Day                                                                                                                                                                                                                                                                         | Task                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                       |       |                                                                                                                                                                                         |                                                                                      |             |
| Sun (7/26)                                                                                                                                                                                                                                                                  | Set up the repo and vite react app with 5 tabs                                                                                         | Sampurn                                                                                                                                                                                                                                                                                                                                                                               | 3     |                                                                                                                                                                                         | Everyone can open it and click the tabs                                              | Completed   |
| Mon (7/27)                                                                                                                                                                                                                                                                  | Auth + hosted DB (sign-up, login, logout, and database-connected), Settings page, Customizing color                                    | Thanhbinh                                                                                                                                                                                                                                                                                                                                                                             | 4     |                                                                                                                                                                                         | Add a task, refresh, it's still there                                                | Completed   |
| Mon                                                                                                                                                                                                                                                                         | Stylesheet from the Figma (colors, fonts, logo, cards), Customizing color palette                                                      | Nami                                                                                                                                                                                                                                                                                                                                                                                  | 3     |                                                                                                                                                                                         | A sample card matches the Figma                                                      | Completed   |
| Mon                                                                                                                                                                                                                                                                         | Vercel connect / Auto deploy git repo                                                                                                  | Sampurn                                                                                                                                                                                                                                                                                                                                                                               | 1     | Repo setup                                                                                                                                                                              | live link                                                                            | Completed   |
| Mon - Tue                                                                                                                                                                                                                                                                   | Google Cloud OAuth setup and Connect Google button: Calendar events + Gmail as tasks                                                   | Madison                                                                                                                                                                                                                                                                                                                                                                               | 4     |                                                                                                                                                                                         | Realistic tasks and events load (start with sample data)                             | Completed   |
| Tue-Wed (7/28-7/29)                                                                                                                                                                                                                                                         | Today screen (greeting, anchors, due today, calendar) + check-off                                                                      | Sampurn                                                                                                                                                                                                                                                                                                                                                                               | 5     |                                                                                                                                                                                         | Looks like the Figma, tapping a circle updates '0 of 3 landed'                       | Completed   |
| Tue-Wed                                                                                                                                                                                                                                                                     | Start Morning briefing (review email/calendar items, add tasks, pick top 3)                                                            | Sampurn                                                                                                                                                                                                                                                                                                                                                                               | 4     | Google cloud OAuth                                                                                                                                                                      | Finishing it fills Today's 3; selecting Gmail shows only gmail tasks                 | Completed   |
| Tue-Wed                                                                                                                                                                                                                                                                     | Reflection screen + rollover tasks (unfinished anchors show up in tomorrow's briefing)                                                 | Sampurn                                                                                                                                                                                                                                                                                                                                                                               | 5     | hosted DB                                                                                                                                                                               | Shows what got done; typing 'essay' suggests a tiny first step                       | Completed   |
| Tue-Wed                                                                                                                                                                                                                                                                     | Focus timer (timer should show up as a popup when leaving screen)                                                                      | Nami                                                                                                                                                                                                                                                                                                                                                                                  | 4     |                                                                                                                                                                                         | Timer runs down and the session is saved                                             | Completed   |
| Mon-Wed                                                                                                                                                                                                                                                                     | This Week screen (Start weekly briefing, 7-day strip, selected-day task and event filtering, task details, empty/loading/error states) | Reed                                                                                                                                                                                                                                                                                                                                                                                  | 5     |                                                                                                                                                                                         | Scrollable week overview, select a day, view that day's tasks and events, open it    | Completed   |
| Thu (7/30)                                                                                                                                                                                                                                                                  | First-step auto-suggest                                                                                                                | Sampurn                                                                                                                                                                                                                                                                                                                                                                               | 2     | today page                                                                                                                                                                              |                                                                                      | Completed   |
| Thu                                                                                                                                                                                                                                                                         | Morning and evening notification settings, reminder scheduling, and test notification                                                  | Reed                                                                                                                                                                                                                                                                                                                                                                                  | 4     |                                                                                                                                                                                         | users can enable notifications, choose briefing and reflection times, receive a test | Completed   |
| Thu                                                                                                                                                                                                                                                                         | Garden screen: plant grows when you land anchors                                                                                       | Sampurn                                                                                                                                                                                                                                                                                                                                                                               | 4     | Base screens setup                                                                                                                                                                      | Landing anchors grows the plant                                                      | Completed   |
| Thu                                                                                                                                                                                                                                                                         | Ambient sound in the timer - White noise / Brown noise / Cafe / Rain sound                                                             | Nami                                                                                                                                                                                                                                                                                                                                                                                  | 3     | Focus timer                                                                                                                                                                             | Sound plays                                                                          | Completed   |
| Thu                                                                                                                                                                                                                                                                         | User manual PDF                                                                                                                        | Thanhbinh                                                                                                                                                                                                                                                                                                                                                                             | 2     | Base screens setup                                                                                                                                                                      | complete user manual                                                                 | Completed   |
| Thu                                                                                                                                                                                                                                                                         | Full run-through of the whole loop                                                                                                     | Madison                                                                                                                                                                                                                                                                                                                                                                               | 1     | All core features complete                                                                                                                                                              | Briefing → timer → check off → reflect works with no dead ends                       | Completed   |
| Thu                                                                                                                                                                                                                                                                         | Gmail tasks filtering [Newly discovered]                                                                                               | Sampurn                                                                                                                                                                                                                                                                                                                                                                               | 4     |                                                                                                                                                                                         |                                                                                      | Completed   |
| Thu                                                                                                                                                                                                                                                                         | dark mode [Newly discovered]                                                                                                           | Sampurn                                                                                                                                                                                                                                                                                                                                                                               | 1     |                                                                                                                                                                                         |                                                                                      | Completed   |
| Thu                                                                                                                                                                                                                                                                         | forgot-password option [Newly discovered]                                                                                              | Sampurn                                                                                                                                                                                                                                                                                                                                                                               | 1     |                                                                                                                                                                                         |                                                                                      | Completed   |
| Thu - Fri (7/30 - 7/31)                                                                                                                                                                                                                                                     | feature testing and stress testing report (2 feature tests + 5 stress tests per person)                                                | Everyone                                                                                                                                                                                                                                                                                                                                                                              | 5     |                                                                                                                                                                                         | complete testing report                                                              | Completed   |
| Fri (7/31)                                                                                                                                                                                                                                                                  | Put the app online, test on a phone, URL of the prototype                                                                              | Sampurn                                                                                                                                                                                                                                                                                                                                                                               | 1     |                                                                                                                                                                                         | Working live URL                                                                     | Completed   |
| Fri                                                                                                                                                                                                                                                                         | Submit implementation document                                                                                                         | Thanhbinh                                                                                                                                                                                                                                                                                                                                                                             | 1     |                                                                                                                                                                                         |                                                                                      | Completed   |
| Fri                                                                                                                                                                                                                                                                         | AI coding tool-use Google form                                                                                                         | Everyone                                                                                                                                                                                                                                                                                                                                                                              | 1     |                                                                                                                                                                                         |                                                                                      | Completed   |
| Anchor Development Plan, Sat 8/1 to Thu 8/6                                                                                                                                                                                                                                 |                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                       |       |                                                                                                                                                                                         |                                                                                      |             |
|                                                                                                                                                                                                                                                                             | 8/1 Better Gmail filter tasks [Newly discovered]                                                                                       | Sampurn                                                                                                                                                                                                                                                                                                                                                                               | 3     |                                                                                                                                                                                         |                                                                                      | Not started |
|                                                                                                                                                                                                                                                                             | 8/1 Sign in with google [Newly discovered]                                                                                             | Sampurn                                                                                                                                                                                                                                                                                                                                                                               | 3     |                                                                                                                                                                                         | Auto morning brieing and notation in meta glasses                                    | Not started |
|                                                                                                                                                                                                                                                                             | 8/1 Ambient sound in the timer with better audio                                                                                       | Nami                                                                                                                                                                                                                                                                                                                                                                                  | 3     |                                                                                                                                                                                         |                                                                                      | Not started |
|                                                                                                                                                                                                                                                                             | 8/1 Verify email creation via signed up email [Newly discovered]                                                                       | Thanhbinh                                                                                                                                                                                                                                                                                                                                                                             | 3     | Working sign up page                                                                                                                                                                    |                                                                                      | Not started |
|                                                                                                                                                                                                                                                                             | 8/2 Delete account [Newly discovered]                                                                                                  | Thanhbinh                                                                                                                                                                                                                                                                                                                                                                             | 3     |                                                                                                                                                                                         | User cannot sign in with their old credentials                                       | Not started |
|                                                                                                                                                                                                                                                                             | 8/2 Password recovery [Newly discovered]                                                                                               | Madison                                                                                                                                                                                                                                                                                                                                                                               | 2     |                                                                                                                                                                                         |                                                                                      | Not started |
|                                                                                                                                                                                                                                                                             | 8/2 Write up protocol                                                                                                                  | Madison                                                                                                                                                                                                                                                                                                                                                                               | 2     |                                                                                                                                                                                         |                                                                                      | Not started |
|                                                                                                                                                                                                                                                                             | 8/2 - 8/6 User Evaluation                                                                                                              | Everyone                                                                                                                                                                                                                                                                                                                                                                              | 10    |                                                                                                                                                                                         |                                                                                      | Not started |
| STRETCH GOALS                                                                                                                                                                                                                                                               |                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                       |       |                                                                                                                                                                                         | Play button reads the plan aloud                                                     |             |
|                                                                                                                                                                                                                                                                             | Loading animation                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                       |       |                                                                                                                                                                                         |                                                                                      |             |
|                                                                                                                                                                                                                                                                             | Spoken morning briefing                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                       |       |                                                                                                                                                                                         |                                                                                      |             |
|                                                                                                                                                                                                                                                                             | Meta Glasses connect                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                       |       |                                                                                                                                                                                         |                                                                                      |             |
|                                                                                                                                                                                                                                                                             | Website version for Laptop/PC                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                       |       |                                                                                                                                                                                         |                                                                                      |             |
| Parallel Work                                                                                                                                                                                                                                                               |                                                                                                                                        | Parallel Work                                                                                                                                                                                                                                                                                                                                                                         |       | Potential Bottleneck                                                                                                                                                                    |                                                                                      |             |
| We split up the work so the individual pages can be done in parallel. So styling, authentication, Google OAuth integration, and development of the Today, This Week, Reflection, and Focus Timer screens can be completed simultaneously after the repository is configured |                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                       |       | Google OAuth and Calendar/Gmail integration are prerequisites for the Morning Briefing feature. We are also worried about finalizing Hi-fi designs before full front-end implementation |                                                                                      |             |
| Outside Constraints                                                                                                                                                                                                                                                         |                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                       |       | Workload Distribution                                                                                                                                                                   |                                                                                      |             |
| Sampurn                                                                                                                                                                                                                                                                     | Summer Research, another summer class                                                                                                  | Workload was determined based on experience and areas of interest for growth. Sam focuses on application architecture and deployment, Madison implements Google integrations and the Morning Briefing experience, Nami develops UI styling and focus features, Thanhbinh works on persistence and the Reflection flow, and Reed develops the weekly planning screen and notifications |       |                                                                                                                                                                                         |                                                                                      |             |
| Madison                                                                                                                                                                                                                                                                     | Summer Research, another summer class                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                       |       |                                                                                                                                                                                         |                                                                                      |             |
| Nami                                                                                                                                                                                                                                                                        | Applying for jobs                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                       |       |                                                                                                                                                                                         |                                                                                      |             |
| Reed                                                                                                                                                                                                                                                                        | 17 units, job, overlapping exams and assignments                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                       |       |                                                                                                                                                                                         |                                                                                      |             |
| Thanhbinh                                                                                                                                                                                                                                                                   | Summer Research, another summer class, co-teaching a program on the week of 8/2 - 8/8                                                  |                                                                                                                                                                                                                                                                                                                                                                                       |       |                                                                                                                                                                                         |                                                                                      |             |
| Notes                                                                                                                                                                                                                                                                       |                                                                                                                                        | git pull before starting and before pushing, git push after finishing<br>everyone commits their own work daily, no single final dump<br>Loading, empty, error, validation, and confirmation states on every screen                                                                                                                                                                    |       |                                                                                                                                                                                         |                                                                                      |             |